KHAN LAB SCHOOL · AI IN EDUCATION SUMMIT 2026

ARENA SESSION · LARGE-GROUP KEYNOTE & CASE STUDIES

When 10-Year-Olds Ship Real Products

AI Projects & Human-Centered PBL



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Who We Are: Rui Li & MarsNavi



Rui Li

Founder, MarsNavi

Beijing, China

- 20 Years in EdTech · 13,000+ Teaching Hours
- Chief Designer: HD Video & Dual-Teacher Live
- Author of “The Core of Product”
- Founded MarsNavi in the AI Era (Ages 8–14)
- 6-Kid Online Teams Solving Real Problems

Session Overview

PART 1

A Solid 7-Step Methodology

Adult Hackathon Opening & 2 Full Online Cases (Bot + Game)

PART 2

Entering the Commercial World

“Friends of Time” NYE Pitch & “Xiao Zhuge” Hardware

PART 3

4 Growth Dimensions & 70-Pt Rule

Logic, Aesthetics, Empathy, Collaboration & Against 1-Click

Opening: 4 Kids at an Adult AI Hackathon

1. Never Learned Coding

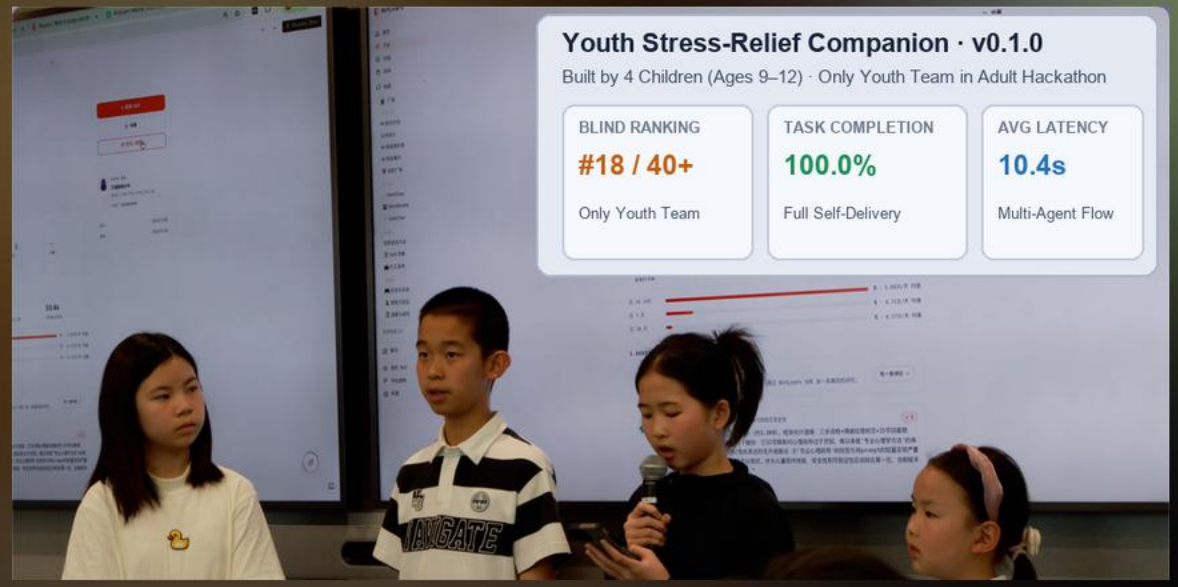
4 kids (ages 9–12) with zero coding competed against 40+ adult teams

2. Relieving Own Stress

Built a Youth Stress-Relief Bot for their own real emotional needs

3. Beat AI Judge · Rank #18

Withstood tough AI-judge critiques on stage and beat 20+ adult teams



#18 Youth Stress-Relief Companion Agent (4 Kids, Ages 9–12) SCORE 52.21
Adult Hackathon (40+ Teams: Engineers, PMs & CS Majors) · Blind AI Evaluation

Part 1 · A Solid 7-Step PBL Methodology



Steps 1–3 · Paper First

Consensus, Best Practices & Sketch

Step 4 · Make It Run

Build Working MVP with AI

Steps 5–7 · Real World

Real Users, Mystery NPC & Ship

Case 1: “Warm Tree-Hole” Emotion Companion Bot

1. Steps 1–2 · Consensus

Scrapped 5 weeks of Dunhuang; asked “What if the user is partly wrong?”

2. Steps 3–4 · Paper & MVP

Drew Emotion Wall on paper; wrote 5 Coze rules to listen before advising

3. Steps 5–7 · Test & Ship

Fixed “Shut up!” bug in peer tests, faced Mom NPC & shipped live



Case 2: “Starfall Route” Card-Shooter Mini-Game

1. Steps 1–3 · Fuse & Sketch

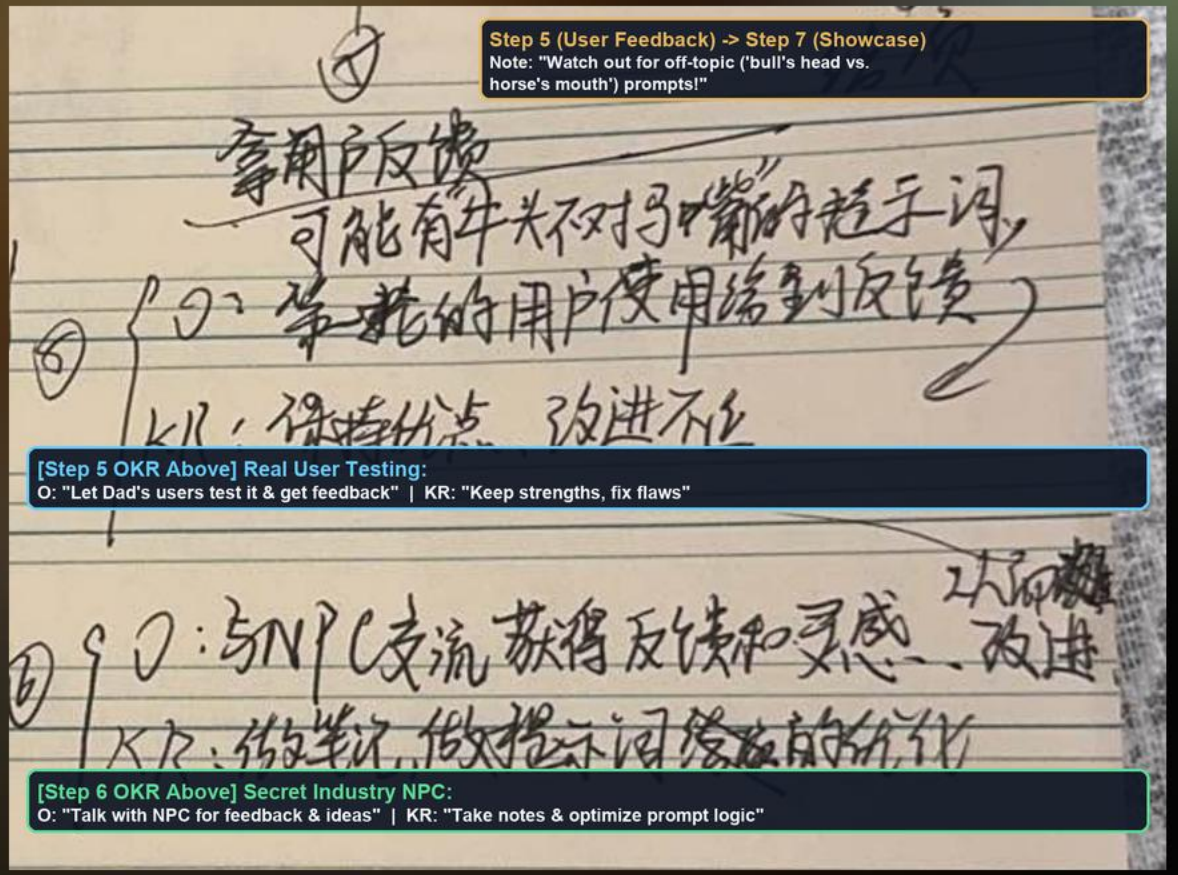
Voted to fuse cards + space shooter;
cut AI's long sci-fi intro essay

2. Step 4 · 50-Level MVP

Coded 3 ships, card combat & shop
in HTML5 with zero coding background

3. Steps 5–7 · Mom's Test

Slowed Lv. 1–10 bullets for Mom;
“Tell AI not to change the fun parts!”



Part 2 · Real Commercial Client: NYE Keynote

1. 1 Month on Ticket Pitch

Designed polygon tickets by user years
+ 2-page handwritten letter & demo

2. Rejected for 3 Reasons

Risk of labeling users, Dec was too
late for tickets & weak demo logic

3. Zero Favors in Business

Coaches kept a poker face in silence —
kids faced real client rejection



给节目组的一封信

尊敬的《跨年演讲》节目组：

你们好！

我是【烛途】AI实践营的张佳齐。我和我的队友，两个想用AI锚定时光的少年。今天，我们深感荣幸能向各位倾诉关于"门票"的思考。

你是否也曾琢磨：一张小小的门票，如何装下这庄重的仪式？当门票有情感，它该如何向每一位观众表达尊重与个性化？

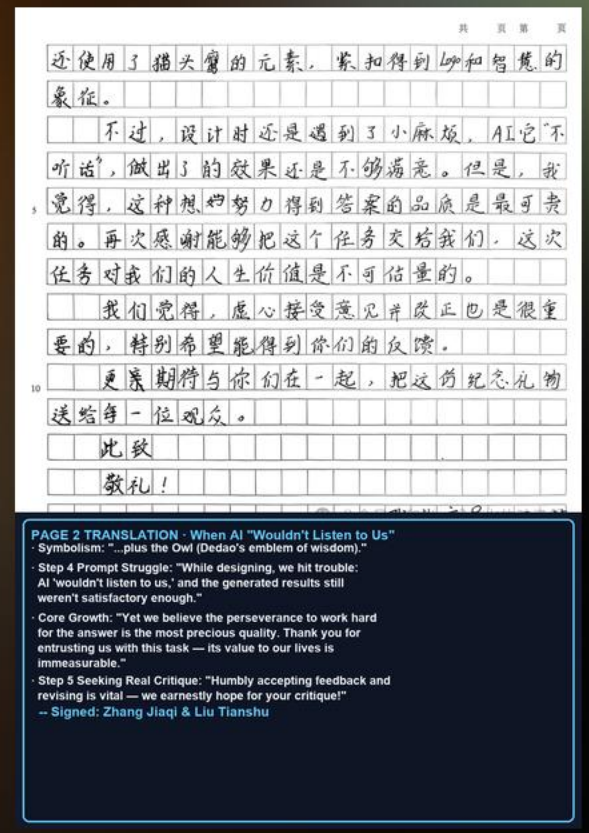
带着这些问题，我们翻找各大平台，搜索了整整二周。翻了上千份设计参考：酒店里温暖的欢迎卡、精美立体的录取通知书、还见有各种重大活动的票根，更反复盯着跨年演讲的海报。把那些让人眼前一亮的东西都挖掘出来。

一张票，不只是进会场的凭证，更是观众对跨年演讲永远的纪念。

所以，我们使用沙漏的元素，让观众感受到时间的流逝，紧扣主题1000天后的世界。我们使用猫头鹰的元素，紧扣主办地三亚的元素。我们

PAGE 1 TRANSLATION · Letter to NYE Keynote Team

- Intro: "I am Zhang Jiaqi from MarsNavi, with my teammate Liu Tianshu."
- Steps 1-2 (3 Wks, 1,000+ Refs): "How can a small ticket hold such a solemn ceremony? For 3 full weeks, we studied 1,000+ references: hotel welcome cards, 3D admission letters, event stubs & NYE posters."
- Step 3 (Design): "Used Hourglass ('World in 1,000 Days') + Sanya elements."



还使用了猫头鹰的元素，紧扣得到Dodo和智慧的象征。

不过，设计时还是遇到了小麻烦，AI它"不听话"，做出的效果还是不够满意。但是，我觉得，这种想办法努力得到答案的品质是最可贵的。再次感谢能够把这个任务交给我们，这次任务对我们的人生价值是不可估量的。

我们觉得，虚心接受意见并改正也是很重要的，特别希望能得到你们的反馈。

更加期待与你们在一起，把这份纪念礼物送给每一位观众。

此致

敬礼！

PAGE 2 TRANSLATION · When AI "Wouldn't Listen to Us"

- Symbolism: "...plus the Owl (Dedao's emblem of wisdom)."
- Step 4 Prompt Struggle: "While designing, we hit trouble: AI 'wouldn't listen to us,' and the generated results still weren't satisfactory enough."
- Core Growth: "Yet we believe the perseverance to work hard for the answer is the most precious quality. Thank you for entrusting us with this task — its value to our lives is immeasurable."
- Step 5 Seeking Real Critique: "Humbly accepting feedback and revising is vital — we earnestly hope for your critique!"

-- Signed: Zhang Jiaqi & Liu Tianshu

NYE Keynote: Agile Pivot & 34 City Covers Live

1. “What Else Can We Do?”

Instead of quitting at age 10, kids asked directors for an urgent task

2. Dropping Sunk Cost

Set aside a month of ticket work for “Calling Out to All Regions” covers

3. 34 Covers on Live TV

Aired to 33M+ viewers — realized “not enough time is a fatal flaw”



“Xiao Zhuge” Story Machine: Skipping Steps 1–3

1. Inside an AI Company

5 kids worked 5 days in Jollybubu
with programmable NFC White Cards

2. Skipping Steps 1–3

Generated 50 Three Kingdoms stories
(10 per kid) in a single afternoon

3. “Running != Finished”

Lengths, styles & settings clashed —
1-click output lacks shared judgment



“Xiao Zhuge”: Back to Paper Design & 3D Print

1. Step 2 · Unique Angle

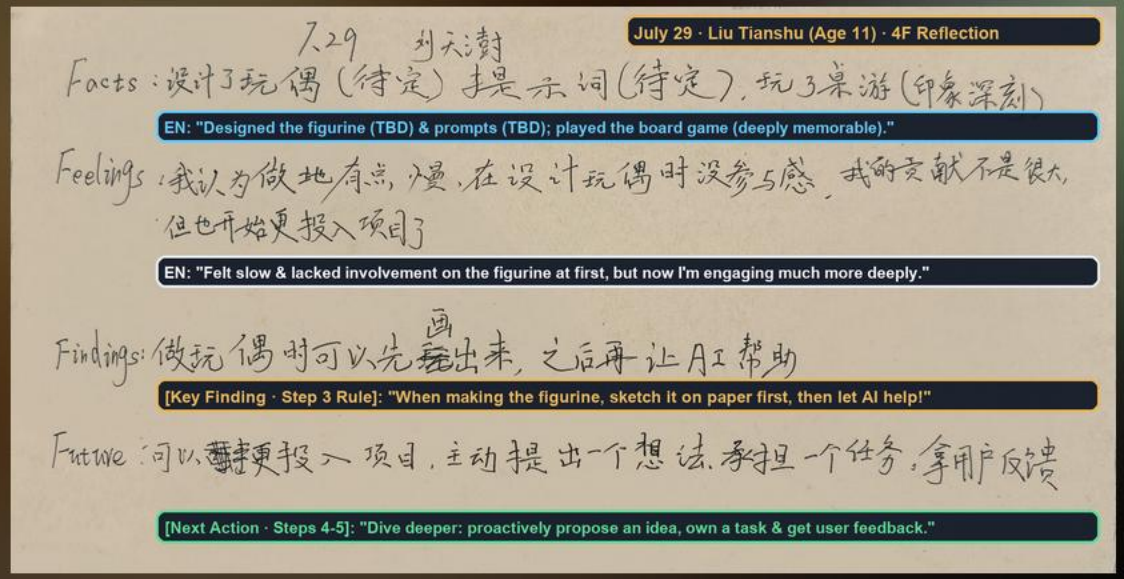
Studied “Uncle Kai Stories” to find a fresh angle over existing audio

2. Step 3 · 1st-Person IP

A kid dreams into Ch. 14 as Zhuge; unified 50 stories + Ep. 51 reflection

3. Step 4 · Paper Before AI

3D-printed NFC figurine; wrote rule: “Sketch on paper first, then use AI”







“Xiao Zhuge”: Park Testing, 0 Sales & Grit

1. Step 5 · Park User Test

Playground kids wanted to run around; polite praise was not real demand



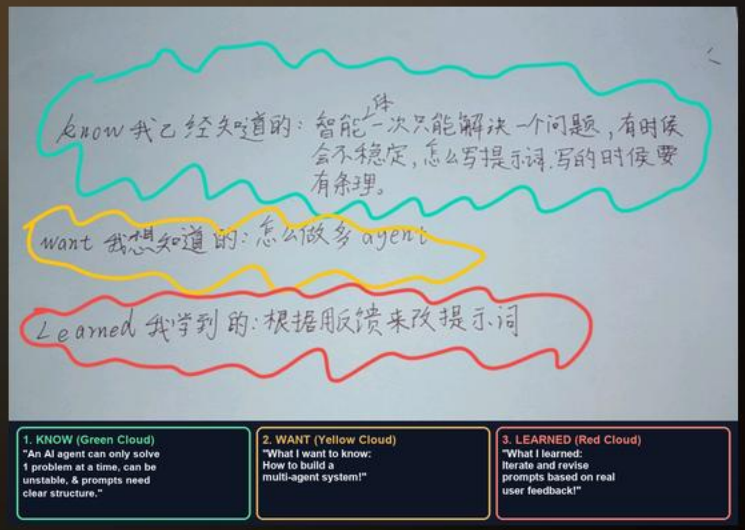
2. “Adults Step Outside”

Asked coaches to leave the room so the 5 kids could reflect alone



3. 0 Sold · Refusing to Quit

Found “fun for kids, useful for moms” and vowed to keep building in fall



Part 3 · Children's Growth Across 4 Dimensions

1. Logic · Clear Structure

From messy wishes to structured prompts

2. Aesthetics · Human Taste

“Not a know-it-all AI” & “It has no soul”

3. Empathy · Real Users

Facing Grandma's & Brother's rejection

4. Collaboration · Partner

Calling on AI when stuck, yet questioning it

VERBATIM CLASSROOM RECORD · YOUTH TREEHOLE

“The AI agent cannot unconditionally take the user’s side — because sometimes the user is also wrong.”

— Guo Mengyao (Age 9)

Recorded during the Youth Emotion Companion project discussion — showing how children debate fairness and boundaries for AI.

AI Capability Is Always the 70-Point Baseline

1. Always 70 Points

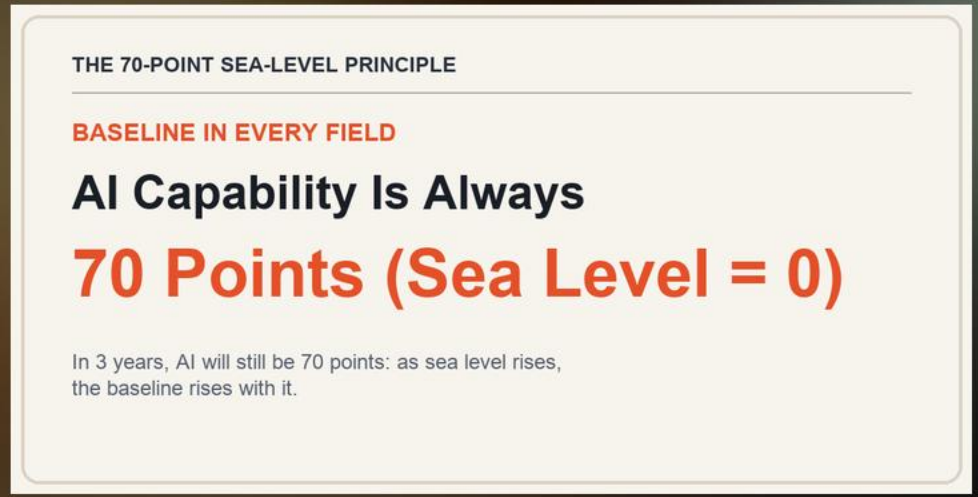
Magical only on what we can't do;
AI is the 70-pt baseline in every field

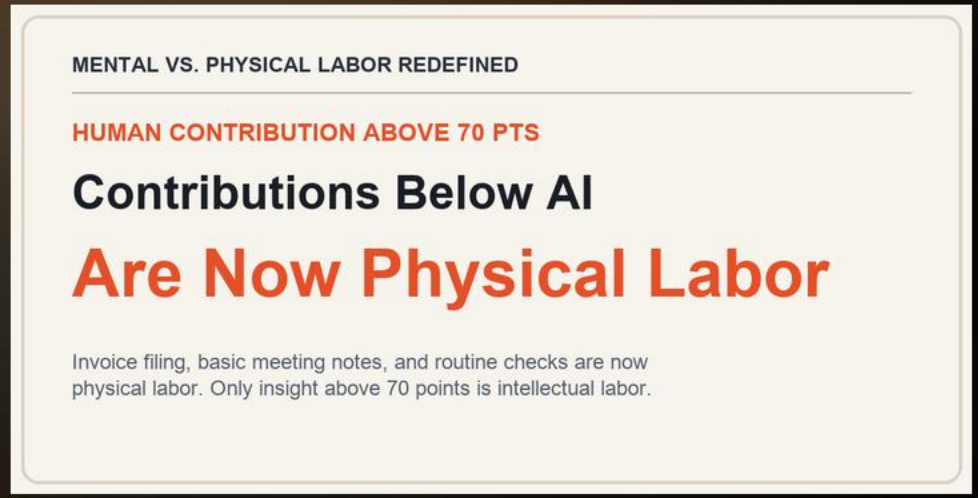
2. Rising 70-Pt Sea Level

As models upgrade, the sea level rises — drowning work below 70 pts

3. Below 70 Pts = Physical

Routine filing & notes are physical labor; only >70-pt insight counts





Against One-Click Generation: Judgment First

1. 2-Sentence Game = 0 Value

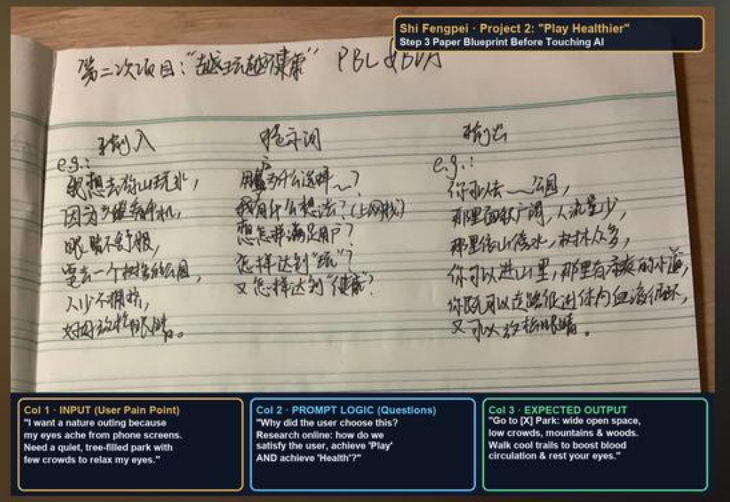
A 2-sentence Minecraft web clone is
<70 pts and puts brains to sleep

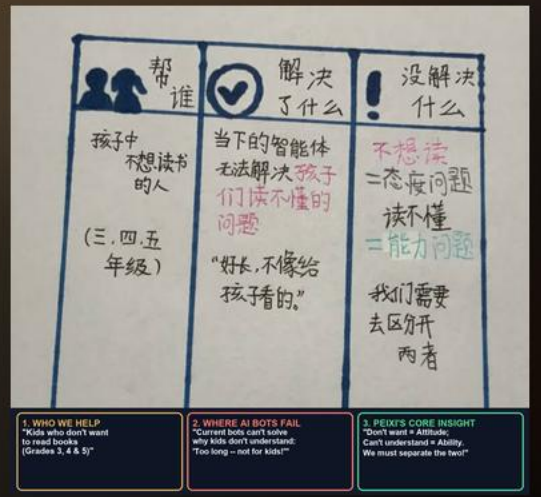
2. Judgment Is Real Thinking

Defining pain points, setting rules
& making trade-offs on paper first

3. Kids Complete Projects

Tools complete generation in 1 click;
only children complete the project





MARSNAVI MISSION · OPEN ACADEMIC & SCHOOL PARTNERSHIPS

Bringing Authentic PBL into Every Child's Learning

Rui Li · Founder of MarsNavi

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